CUSUM Active-Manager Monitor

Statistical Process Control for Portfolio Manager Oversight

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What is CUSUM?

CUSUM (Cumulative Sum) is a sequential change-point detection method from statistical process control, adapted for investment management by Philips, Yashchin & Stein (2003). It answers one critical question:

Has a fund manager lost their investment skill — and if so, exactly when?

The Case for CUSUM — Why CIOs Should Care

Cost of Inaction:

Retaining an underperforming manager for 3 extra years on a 500M mandate at IR=-0.3, TE=6% costs ~9M in excess losses. Early, systematic detection pays.

Faster than Rolling IR:

A trailing 3-year IR review needs 10+ years of data to distinguish skill from luck at 95% confidence. CUSUM detects persistent CHANGES in IR within 2-3 years.

Rigorous False-Alarm Control:

Threshold h is calibrated from the paper's ARL table: h=19.81 gives ~1 false alarm per 60 months (5 years) when the true IR=+0.5. Fewer spurious terminations.

Two-Sided Monitoring:

Detects skill LOSS (lower CUSUM) and unexpected skill GAIN (upper CUSUM), enabling both firing decisions and early-stage manager promotions.

Rolling Restart:

After each alarm L resets to zero — monitoring never stops. Multiple alarms build a richer signal; post-alarm returns can be tracked to validate each flag.

Industry Precedent:

SPC methods are standard in manufacturing, clinical trials, and nuclear operations. Mercer and WTW publish SPC-adjacent monitoring frameworks for manager oversight.

Actionable, Not Automatic:

An alarm triggers a structured review — not automatic termination. It forces the investment committee to ask: Is the alpha source still intact?

Reference: Philips, T., Yashchin, E. & Stein, D. (2003). "Using Statistical Process Control to Monitor Active Managers." Journal of Portfolio Management, 30(1), 86-94.

How CUSUM Works

Step-by-step mechanics of the Philips-Yashchin-Stein procedure

1

Step 1 — Log Excess Return

```
Each month:   et = ln[(1 + r_fund) / (1 + r_benchmark)]
```

Log returns are additive and symmetric — they handle compounding correctly.

2

Step 2 — EWMA Tracking-Error Estimate

```
sigma^2_t = gamma * sigma^2_{t-1} + (1-gamma) * 0.5*(et - e_{t-1})^2   [gamma=0.90]
```

Von Neumann adjacent-difference estimator: robust to drift in the mean, adapts as the manager's risk budget changes.

3

Step 3 — Monthly Information Ratio Estimate

```
IR_hat_t = (et * sqrt(12)) / sigma_{t-1}
```

Annualised IR using lagged sigma to keep the estimator approximately unbiased.

4

Step 4 — Lindley Recursion (Page's CUSUM)

```
L_t = max(0, L_{t-1} + drift)    drift = 0.5*(IR_good + IR_bad) - IR_hat_t
```

Default: IR_good=0.5, IR_bad=0.0 -> drift = 0.25 - IR_hat. L grows when manager underperforms, resets to 0 when they outperform. Upper CUSUM L_up uses drift_up = IR_hat - 0.5*(IR_exceptional + IR_good).

5

Step 5 — Rolling Alarm Rule

```
ALARM if L_t >= h   [default h = 19.81]   then reset L_t = 0 and continue
```

After each alarm L resets to zero — monitoring never stops. All alarm times are recorded; post-alarm returns validate each signal.

Alarm Threshold Reference Table (Table 2, Philips et al. 2003)

Threshold (h)	ARL — Good (IR=0.5)	ARL — Flat (IR=0)	ARL — Bad (IR=-0.5)
11.81	24.00	16.00	11.00
15.00	36.00	22.00	15.00
17.60	48.00	27.00	18.00
19.81	**60.00**	**32.00**	**21.00** *<- default*
21.79	72.00	37.00	23.00
23.59	84.00	41.00	25.00

Fund Universe

41 active managers — page 1 of 3

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
1	FCNTX	Fidelity Contrafund vs Russell 1000 Growth	IWF	US Large Growth	Jan 2005	19.81
2	DODGX	Dodge & Cox Stock vs Russell 1000 Value	IWD	US Large Value	Jan 2005	19.81
3	AGTHX	AmFunds Growth Fund of America vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
4	OTCFX	T. Rowe Price Small-Cap Stock vs Russell 2000	IWM	US Small Cap	Jan 2005	19.81
5	OAKIX	Oakmark International vs MSCI EAFE	EFA	Intl Developed	Jan 2005	19.81
6	TEDMX	Templeton Developing Markets vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
7	VWELX	Vanguard Wellington vs 60/40 Allocation	AOR	Balanced 60/40	Jan 2005	19.81
8	PTTRX	PIMCO Total Return vs US Agg Bond	AGG	US Fixed Income	Jan 2005	19.81
9	PRWCX	T. Rowe Price Capital Appreciation vs 60/40 Allocation	AOR	Balanced 60/40	Jan 2005	19.81
10	FMAGX	Fidelity Magellan vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
11	SEQUX	Sequoia Fund vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
12	FLPSX	Fidelity Low-Priced Stock vs Russell 2000	IWM	US Small/Mid Cap	Jan 2005	19.81
13	DODWX	Dodge & Cox International vs MSCI EAFE	EFA	Intl Value	Jan 2005	19.81
14	LSBRX	Loomis Sayles Bond vs US Agg Bond	AGG	US Fixed Income	Jan 2005	19.81
15	NWFFX	New World Fund vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
16	PRMSX	T. Rowe Price EM Stock vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
17	JEMWX	JPMorgan EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
18	ODMAX	Invesco Developing Markets vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
19	PEIFX	PGIM Jennison EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
20	LZOEX	Lazard EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81

US Equity (FCNTX, DODGX, AGTHX, OTCFX, FMAGX, SEQUX, FLPSX)
Intl & Emerging Markets (OAKIX, TEDMX, DODWX, + 27 EM funds)
Fixed Income / Balanced (VWELX, PTTRX, PRWCX, LSBRX)

Fund Universe

41 active managers — page 2 of 3

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
21	HLMEX	Harding Loevner EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
22	MEMCX	Matthews EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
23	VEIEX	Vanguard EM Stock Index vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
24	VEMAX	Vanguard EM Stock Index Admiral vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
25	NEWFX	American Funds New World vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
26	HAEMX	Harbor EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
27	TEMFX	Templeton EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
28	SFENX	Seafarer EM Growth & Income vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
29	REMIX	Royce EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
30	ABEMX	AB EM Growth vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
31	GERIX	Goldman Sachs EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
32	HLEMX	Harding Loevner EM Institutional vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
33	BGEFX	Baron EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
34	WIEMX	William Blair EM Growth vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
35	WAEMX	Western Asset EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
36	MIEMX	MFS EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
37	EEMAX	Eaton Vance EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
38	FSEAX	Fidelity EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
39	LZOXX	Lazard EM Institutional vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
40	MADCX	Madison EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81

Fund Universe

41 active managers — page 3 of 3

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
41	PDEZX	PGIM EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81

Summary Results

CUSUM alarm status and key statistics (Jan 2005 - present) — page 1 of 3

Fund	Bench	Months	Ann TE	IR (95% CI)	Cum Excess	# Alarms	Alarm?	First Alarm
FCNTX	IWF	256	4.3%	**-0.00 [-0.39,+0.38]**	+3.4%	9	**YES**	**Feb 2010**
DODGX	IWD	256	4.4%	+0.21 [-0.24,+0.67]	+14.5%	7	**YES**	**Mar 2008**
AGTHX	SPY	256	4.6%	+0.13 [-0.33,+0.63]	+9.9%	9	**YES**	**Oct 2008**
OTCFX	IWM	256	4.1%	+0.24 [-0.22,+0.75]	+32.7%	6	**YES**	**Nov 2007**
OAKIX	EFA	256	8.2%	+0.14 [-0.30,+0.57]	+11.0%	8	**YES**	**Sep 2007**
TEDMX	EEM	256	5.9%	**+0.03 [-0.31,+0.41]**	+2.7%	5	**YES**	**Jan 2007**
VWELX	AOR	210	4.0%	**+0.65 [+0.39,+1.10]**	+58.0%	0	**NO**	**-**
PTTRX	AGG	256	1.9%	**+0.50 [+0.03,+1.03]**	+22.2%	5	**YES**	**Dec 2008**
PRWCX	AOR	210	4.6%	**+0.99 [+0.67,+1.45]**	+119.1%	1	**YES**	**Jan 2026**
FMAGX	SPY	256	5.7%	**-0.15 [-0.66,+0.40]**	-24.0%	9	**YES**	**Sep 2006**
SEQUX	SPY	256	8.4%	**-0.22 [-0.64,+0.23]**	-33.2%	9	**YES**	**Apr 2007**
FLPSX	IWM	256	8.3%	**+0.00 [-0.38,+0.37]**	+17.2%	6	**YES**	**Aug 2008**
DODWX	EFA	216	7.1%	**+0.48 [+0.05,+0.96]**	+72.3%	3	**YES**	**Jan 2016**
LSBRX	AGG	256	6.9%	+0.26 [-0.25,+0.88]	+39.1%	7	**YES**	**Jul 2008**
NWFFX	EEM	256	7.8%	**+0.06 [-0.30,+0.43]**	+29.7%	7	**YES**	**Jan 2006**
PRMSX	EEM	256	4.6%	**-0.06 [-0.56,+0.41]**	-9.6%	6	**YES**	**Sep 2008**
JEMWX	EEM	149	5.0%	**+0.42 [-0.21,+1.15]**	+27.3%	2	**YES**	**Jan 2022**
ODMAX	EEM	256	5.9%	**+0.06 [-0.33,+0.43]**	+14.2%	5	**YES**	**Apr 2015**

Summary Results

CUSUM alarm status and key statistics (Jan 2005 - present) — page 2 of 3

Fund	Bench	Months	Ann TE	IR (95% CI)	Cum Excess	# Alarms	Alarm?	First Alarm
PEIFX	EEM	131	8.6%	+0.25 [-0.47,+0.91]	+22.9%	4	**YES**	**Apr 2019**
LZOEX	EEM	256	6.3%	+0.14 [-0.29,+0.56]	+23.4%	4	**YES**	**Mar 2015**
HLMEX	EEM	247	5.7%	**-0.09 [-0.45,+0.30]**	-7.1%	6	**YES**	**Oct 2008**
MEMCX	EEM	256	4.6%	**-0.36 [-0.72,-0.04]**	-26.9%	8	**YES**	**Sep 2007**
VEIEX	EEM	256	4.1%	**-0.12 [-0.46,+0.29]**	-6.8%	5	**YES**	**Oct 2008**
VEMAX	EEM	239	3.7%	**-0.06 [-0.44,+0.35]**	-1.5%	5	**YES**	**Oct 2008**
NEWFX	EEM	256	7.8%	**+0.06 [-0.30,+0.43]**	+29.4%	7	**YES**	**Jan 2006**
HAEMX	EEM	47	20.8%	**-0.49 [-1.03,+1.09]**	-39.3%	2	**YES**	**Mar 2020**
TEMFX	EEM	256	10.5%	**-0.23 [-0.68,+0.19]**	-33.1%	9	**YES**	**Jan 2006**
SFENX	EEM	220	6.6%	**+0.08 [-0.39,+0.54]**	+10.0%	6	**YES**	**Jun 2012**
REMIX	EEM	76	17.5%	**+0.04 [-0.74,+0.84]**	+13.1%	2	**YES**	**Mar 2025**
ABEMX	EEM	228	5.9%	+0.20 [-0.24,+0.65]	+30.8%	5	**YES**	**Dec 2013**
GERIX	EEM	223	4.5%	+0.27 [-0.15,+0.67]	+21.5%	3	**YES**	**Oct 2008**
HLEMX	EEM	83	5.7%	**-0.38 [-1.02,+0.36]**	-15.7%	3	**YES**	**Sep 2020**
BGEFX	EEM	133	12.2%	**+0.10 [-0.58,+0.80]**	+6.9%	5	**YES**	**Aug 2016**
WIEMX	EEM	123	12.0%	**-0.06 [-0.65,+0.57]**	-10.2%	5	**YES**	**Jan 2017**
WAEMX	EEM	224	11.3%	+0.18 [-0.28,+0.63]	+40.9%	7	**YES**	**Oct 2008**
MIEMX	EEM	158	7.2%	**-0.03 [-0.50,+0.45]**	+2.3%	5	**YES**	**Apr 2011**

Summary Results

CUSUM alarm status and key statistics (Jan 2005 - present) — page 3 of 3

Fund	Bench	Months	Ann TE	IR (95% CI)	Cum Excess	# Alarms	Alarm?	First Alarm
EEMAX	EEM	224	5.5%	**+0.08 [-0.42,+0.59]**	+4.7%	5	**YES**	**Sep 2008**
FSEAX	EEM	256	8.9%	**+0.41 [-0.08,+0.93]**	+120.3%	4	**YES**	**Apr 2008**
MADCX	EEM	256	5.3%	+0.13 [-0.22,+0.49]	+14.7%	5	**YES**	**Sep 2008**
PDEZX	EEM	140	11.1%	+0.23 [-0.41,+0.86]	+27.6%	4	**YES**	**Mar 2017**

Post-Alarm Analysis

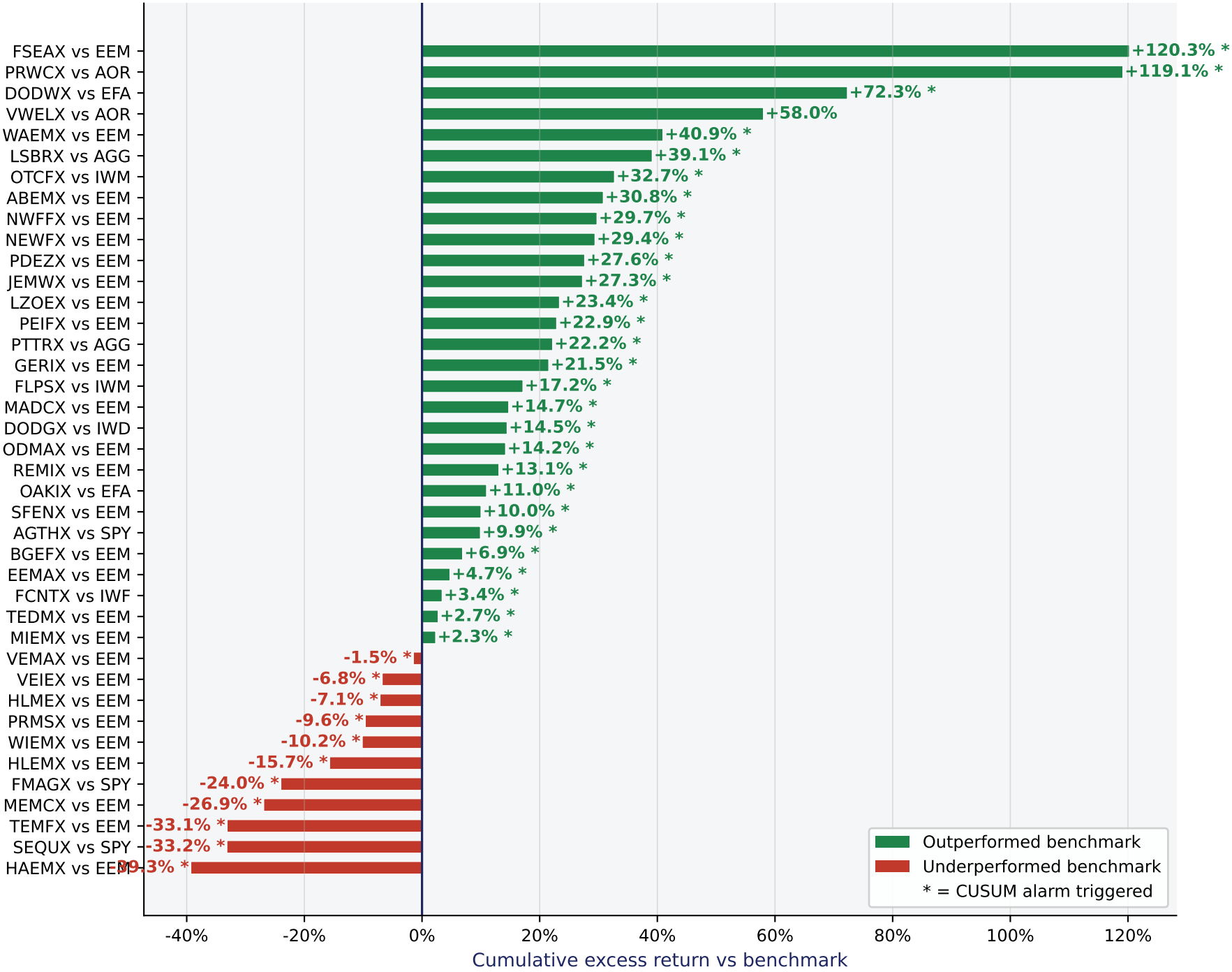
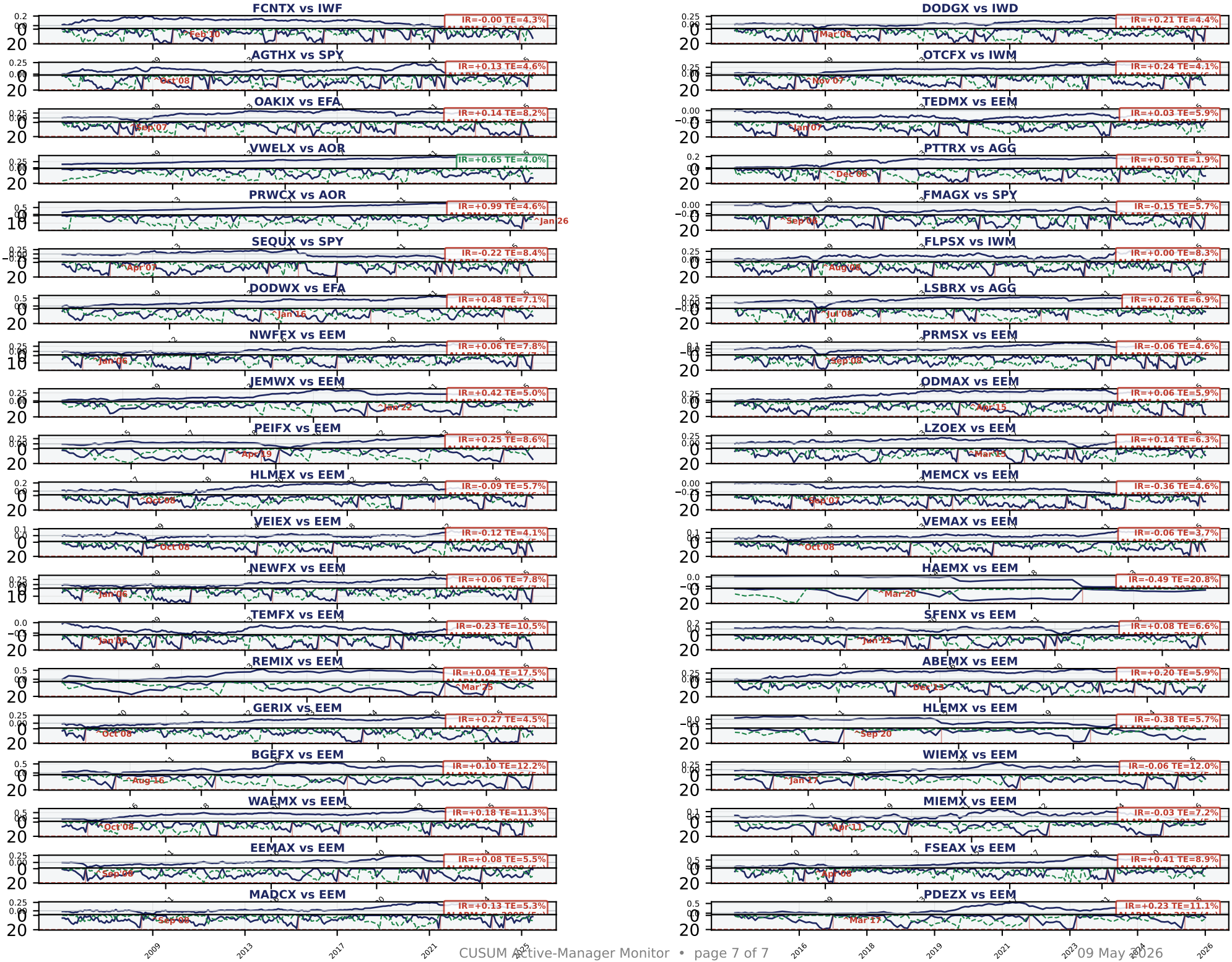
Cumulative excess return vs benchmark in the months following the FIRST alarm

Fund	First Alarm	+12M Fund	+12M Bench	+12M Excess	+24M Excess	+36M Excess	12M Signal
FCNTX	Feb 2010	+24.8%	+24.9%	-0.1%	-2.1%	-0.8%	CORRECT
DODGX	Mar 2008	-45.2%	-42.7%	-2.5%	+2.4%	+0.7%	CORRECT
AGTHX	Oct 2008	+15.6%	+9.7%	+5.9%	+5.0%	-2.1%	FALSE
OTCFX	Nov 2007	-37.5%	-37.6%	+0.0%	+7.5%	+14.0%	FALSE
OAKIX	Sep 2007	-28.9%	-28.8%	-0.1%	+11.9%	+20.3%	CORRECT
TEDMX	Jan 2007	+10.1%	+21.3%	-11.2%	-7.2%	-9.1%	CORRECT
PTTRX	Dec 2008	+13.9%	+3.0%	+10.9%	+14.1%	+10.9%	FALSE
PRWCX	Jan 2026	n/a	n/a	n/a	n/a	n/a	FALSE
FMAGX	Sep 2006	+22.8%	+16.4%	+6.4%	-3.7%	+1.3%	FALSE
SEQUX	Apr 2007	+3.1%	-5.0%	+8.1%	+18.2%	+12.9%	FALSE
FLPSX	Aug 2008	-11.0%	-21.3%	+10.3%	+12.8%	+14.2%	FALSE
DODWX	Jan 2016	+31.9%	+10.8%	+21.1%	+22.8%	+23.0%	FALSE
LSBRX	Jul 2008	-0.8%	+7.3%	-8.0%	+1.6%	+12.6%	CORRECT
NWFFX	Jan 2006	+24.2%	+15.0%	+9.1%	+11.9%	+10.8%	FALSE
PRMSX	Sep 2008	+12.8%	+16.4%	-3.6%	+0.4%	+3.2%	CORRECT
JEMWX	Jan 2022	-13.9%	-13.3%	-0.6%	-2.3%	-4.4%	CORRECT
ODMAX	Apr 2015	-14.1%	-17.9%	+3.8%	+4.5%	+6.0%	FALSE
PEIFX	Apr 2019	-23.0%	-14.2%	-8.8%	-4.4%	+9.4%	CORRECT
LZOEX	Mar 2015	-11.7%	-12.7%	+0.9%	+5.4%	-0.8%	FALSE
HLMEX	Oct 2008	+48.9%	+51.0%	-2.1%	+0.4%	+0.9%	CORRECT
MEMCX	Sep 2007	-33.7%	-29.7%	-3.9%	-8.5%	-7.4%	CORRECT
VEIEX	Oct 2008	+56.0%	+51.0%	+4.9%	+4.7%	+5.6%	FALSE
VEMAX	Oct 2008	+60.2%	+51.0%	+9.2%	+12.1%	+12.5%	FALSE
NEWFX	Jan 2006	+24.2%	+15.0%	+9.1%	+11.9%	+10.8%	FALSE
HAEMX	Mar 2020	+35.1%	+58.8%	-23.7%	-54.4%	n/a	CORRECT
TEMFX	Jan 2006	+15.0%	+15.0%	-0.0%	-18.9%	-5.6%	CORRECT
SFENX	Jun 2012	+2.3%	+0.3%	+2.0%	+5.0%	-2.7%	FALSE
REMIX	Mar 2025	+16.8%	+33.1%	-16.3%	n/a	n/a	CORRECT
ABEMX	Dec 2013	-2.4%	-3.9%	+1.5%	+3.6%	+4.9%	FALSE
GERIX	Oct 2008	+62.3%	+51.0%	+11.3%	+17.2%	+19.0%	FALSE
HLEMX	Sep 2020	+21.0%	+16.0%	+5.0%	-4.8%	-5.5%	FALSE
BGEFX	Aug 2016	+45.1%	+24.7%	+20.4%	+38.9%	+36.6%	FALSE
WIEMX	Jan 2017	+36.2%	+39.4%	-3.2%	-8.3%	+21.2%	CORRECT
WAEMX	Oct 2008	+83.0%	+51.0%	+31.9%	+97.5%	+103.4%	FALSE
MIEMX	Apr 2011	-14.5%	-13.9%	-0.6%	+3.1%	-0.2%	CORRECT
EEMAX	Sep 2008	+15.7%	+16.4%	-0.7%	+6.0%	+6.9%	CORRECT
FSEAX	Apr 2008	-42.3%	-39.8%	-2.5%	-9.7%	-6.5%	CORRECT
MADCX	Sep 2008	+23.3%	+16.4%	+6.9%	+5.5%	+9.9%	FALSE
PDEZX	Mar 2017	+33.8%	+25.0%	+8.8%	+10.1%	+21.2%	FALSE

No alarm (no post-alarm analysis): VWELX

Cumulative Excess Returns — Ranked

Total return of each fund vs its benchmark (Jan 2005 - present)



Individual Fund CUSUM Charts